

NTIRE 2026 Challenge on Robust AI-Generated Image Detection in the Wild

Aleksandr Gushchin* Khaled Abud Ekaterina Shumitskaya Artem Filippov
 Georgii Bychkov Sergey Lavrushkin Mikhail Erofeev Anastasia Antsiferova
 Changsheng Chen Shunquan Tan Radu Timofte Dmitry Vatolin Chuanbiao Song
 Zijian Yu Hao Tan Jun Lan Zhiqiang Yang Yongwei Tang Zhiqiang Wu
 Jia Wen Seow Hong Vin Koay Haodong Ren Feng Xu Shuai Chen
 Ruiyang Xia Qi Zhang Yaowen Xu Zhaofan Zou Hao Sun Dagong Lu
 Mufeng Yao Xinlei Xu Fei Wu Fengjun Guo Cong Luo Hardik Sharma
 Aashish Negi Prateek Shaily Jayant Kumar Sachin Chaudhary Akshay Dudhane
 Praful Hambarde Amit Shukla Zhilin Tu Fengpeng Li Jiamin Zhang
 Jianwei Fei Kemou Li Haiwei Wu Bilel Benjdira Anas M. Ali Wadii Boulila
 Chenfan Qu Junchi Li
 alexander.gushchin@graphics.cs.msu.ru

Abstract

This paper presents an overview of the NTIRE 2026 Challenge on Robust AI-Generated Image Detection in the Wild, held in conjunction with the NTIRE workshop at CVPR 2026. The goal of this challenge was to develop detection models capable of distinguishing real images from generated ones in realistic scenarios: the images are often transformed (cropped, resized, compressed, blurred) for practical usage, and therefore, the detection models should be robust to such transformations. The challenge is based on a novel dataset consisting of 108,750 real and 185,750 AI-generated images from 42 generators comprising a large variety of open-source and closed-source models of various architectures, augmented with 36 image transformations. Methods were evaluated using ROC AUC on the full test set, including both transformed and untransformed images. A total of 511 participants registered, with 20 teams submitting valid final solutions. This report provides a comprehensive overview of the challenge, describes the proposed solutions, and can be used as a valuable reference for researchers and practitioners in increasing the robustness of the detection models to real-world transformations.

1. Introduction

The emergence of generative AI has made the synthesis of photorealistic images increasingly accessible. Modern generative models — including diffusion-based architectures [64], generative adversarial networks [22], and autoregressive image generators [39] — are capable of producing images that are visually indistinguishable from real photographs. The widespread availability of such models through consumer-facing applications has led to an unprecedented volume of AI-generated imagery in circulation, raising serious concerns in the areas of media integrity, misinformation, digital forensics, and content authentication.

In recent years, the research community has developed a variety of methods for detecting AI-generated images. Early approaches [78] exploited generator-specific artifacts such as spectral periodicities characteristic of GAN up-sampling [19] and checkerboard patterns introduced by transposed convolutions [71]. More recent methods target the residual artifacts of diffusion model denoising pipelines [81], or leverage large pretrained vision encoders fine-tuned on real/fake corpora [55].

Several prior works and competitions have contributed datasets and benchmarks for AI-generated image detection [30, 87, 92]. Despite notable progress, a fundamental and persistent limitation of existing detectors is their lack of robustness to real-world image transformations. In practical deployment scenarios, images are routinely cropped, resized, recompressed, and blurred before they are encountered by a detector — operations that can substantially degrade detection performance [37]. However, existing benchmarks typically either cover a limited number of gen-

*A. Gushchin (alexander.gushchin@graphics.cs.msu.ru), K. Abud, E. Shumitskaya, A. Filippov, G. Bychkov, M. Erofeev, S. Lavrushkin, M. Erofeev, A. Antsiferova, D. Vatolin, C. Chen, S. Tan and R. Timofte were the challenge organizers, while the other authors participated in the challenge. Sec. 7 contains the authors' teams and affiliations. NTIRE 2026 webpage: <https://cvlai.net/ntire/2026/>

erators or do not systematically account for real-world image transformations, or evaluate both aspects in isolation. Therefore, the creation of modern detectors that are robust to such transformations remains a serious gap in AI-generated image detection.

This paper presents the NTIRE 2026 Challenge on Robust AI-Generated Image Detection in the Wild, which aims to explore and enhance the robustness of detectors to real-world image transformations, including operations that can substantially disrupt detectors trained on clean data. The challenge is built around a novel large-scale dataset comprising 108,750 real images and 185,750 AI-generated images sourced from 42 generators that span a wide range of open-source and closed-source architectures. Crucially, the dataset incorporates 36 image transformation types reflecting realistic post-processing and distribution conditions, enabling a principled assessment of detector robustness.

This paper provides an overview of the methods submitted to the challenge and reports on their performance. The ideas proposed in this challenge can be used to enhance the robustness of the detectors to real-world transformations, in addition to increasing the detection accuracy that is important in critical spheres.

This challenge is one of the challenges associated with the NTIRE 2026 Workshop¹ on: deepfake detection [31], high-resolution depth [89], multi-exposure image fusion [61], AI flash portrait [26], professional image quality assessment [59], light field super-resolution [80], 3D content super-resolution [77], bitstream-corrupted video restoration [93], X-AIGC quality assessment [44], shadow removal [74], ambient lighting normalization [73], controllable Bokeh rendering [65], rip current detection and segmentation [16], low light image enhancement [12], high FPS video frame interpolation [13], Night-time dehazing [2, 3], learned ISP with unpaired data [58], short-form UGC video restoration [40], raindrop removal for dual-focused images [41], image super-resolution (x4) [10], photography retouching transfer [17], mobile real-world super-resolution [38], remote sensing infrared super-resolution [42], AI-Generated image detection [27], cross-domain few-shot object detection [60], financial receipt restoration and reasoning [25], real-world face restoration [76], reflection removal [6], anomaly detection of face enhancement [91], video saliency prediction [49], efficient super-resolution [63], 3d restoration and reconstruction in adverse conditions [43], image denoising [67], blind computational aberration correction [69], event-based image deblurring [68], efficient burst HDR and restoration [57], low-light enhancement: ‘twilight cowboy’ [34], and efficient low light image enhancement [85].

¹<https://www.cvlai.net/ntire/2026/>

2. Challenge

Our challenge is dedicated to the task of distinguishing AI-generated images produced in Text-to-Image setting from real imagery. We focus on the most general image domain, without confining to particular subdomains such as faces, humans, or specific object classes. The goals of this competition is threefold: 1) Assess the current state of AI-powered image generation and its differentiability from the real images; 2) Evaluate the detection robustness under complex image degradation pipelines; 3) Appraise the generalization capabilities of the modern detectors to unseen generators.

2.1. Dataset

For this challenge, we introduce our novel dataset containing both high-quality real-world imagery and AI-generated images sourced from a diverse set of generators, ranging from older Stable Diffusion [64] models up to the most recent models to date (e.g., Nano Banana 2 [53], SeeDream 5 Lite [54]).

Real subset. To construct a diverse and comprehensive set of real images that represent “in-the-wild” content, we collect them from 3 large-scale web-sourced image-text datasets: CC12M [9], CommonPool [21] and RedCaps [15]. After initial filtering for explicit and inappropriate content and data availability check, a total of ~ 12 M images were selected. To increase data quality, we employed series of additional filtering stages, ranging from resolution thresholding and CLIP-deduplication to complex VLM-based image categorization and scoring, further reducing the number of suitable images by 90%. Finally, we sample 100,000 unique real images from our filtered set for the training stage, and ~ 9 k in total for validation and test splits.

Generated subset. To produce generated images, we utilize 42 open-source and proprietary Text-to-Image generators released between 2022 and 2026. Generation prompts for this subset are collected from the corresponding real images: we first employ a Large Vision-Language model to produce detailed image captions and then rewrite them into concise and structured prompts using LLM. By “pairing” generated images with their real counterparts, we ensure that both subsets reflect similar semantics and content distribution, which should help detectors learn content-agnostic features. To further minimize potential biases in generated imagery, we also align its distributions of resolutions, aspect ratios, JPEG compression quality factors, and other statistics to those of the real subset. For the training split, for each real image (out of 100k), we randomly select from 1 to 3 generators to produce corresponding generated images, resulting in ~ 177 k samples. The training split covers 20 open-source generators in total, with the most recent models including Flux Kontext (dev) [36], DeepFloyd-IF [50], and Ovis-image [75]. Most of the top-performing open-source models (e.g. Qwen-Image [83], HiDream [7], etc.)

as well as proprietary generators (Nano Banana [52], Grok Imagine [51], etc.) were reserved for validation and test splits, with each consecutive split containing progressively larger share of state-of-the-art models. For validation and test splits, we use only unique images without its paired counterpart to avoid potential advantage from selecting between multiple similar images. More details on the contents of each split can be found in Appendix in Tables 3 and 4.

Robust detection track. Our competition includes a special track designed to evaluate detectors’ performance in a more challenging scenario where input images might be significantly distorted to conceal its nature. This track reflects various image corruptions and artifacts which might be encountered in real-world detector deployment. To this purpose, we employed an image degradation pipeline inspired by [1]. Each image is transformed with 1 to 5 randomly sampled consecutive distortions from different groups (e.g., noise, compression, blur, etc.). Each transformation has multiple magnitude levels, which are sampled independently on each application. In the robust detection track, both real and generated images undergo the same degradation pipeline. To avoid detector tuning for specific distortion types, we employed different sets of distortions in each dataset split and progressively complicated them in the last stages of the challenge. To identify the most challenging distortion types, we utilized our pre-trained detectors and analyzed their performance on degraded images. In total, we employ 36 different transformation types across the challenge stages, ranging from simple transforms like Gaussian blur and white noise, up to complex techniques like invisible image watermarking [84], neural compression [4, 11] and watermark-erasing adversarial attacks [66]. Full list of the transformations can be found in Tables 3 and 4. In validation and test splits, half of the real and generated images are selected for the robust track and distorted with the aforementioned pipeline.

2.2. Evaluation metrics

- Primary metric – Robust ROC AUC, computed over all distorted images in the set. It measures the detector’s global discriminative ability under complex postprocessing pipelines.
- Secondary metric – Clean ROC AUC, computed over all non-distorted images in the set. It measures detector performance in a clear setting, disregarding potential image transformations.

ROC AUC is measured between binary labels (0/1), where 1 indicates a fake image, and the confidence scores submitted by the participants. We selected ROC AUC over Accuracy or F1 score, as it does not require quantizing participants’ predictions to binary scores with a predefined threshold.

3. Challenge Methods and Teams

3.1. MICV

3.1.1. Technical Details

Overview In this work, we propose a robust ensemble-based framework for AI-generated image detection, specifically designed to address the challenges of domain generalization and cross-platform detection. Our approach centers on three core pillars:

- **Hierarchical Data Strategy:** We curate a large-scale, multi-source training corpus that integrates open-source academic benchmarks, synthetic images from cutting-edge generative models, and high-fidelity samples from closed-source commercial APIs, ensuring comprehensive coverage of diverse generative artifacts.
- **Ensemble-based Architecture:** We utilize the powerful representation capabilities of multiple DINOv3 backbones, organized into two distinct model committees. By employing a late-fusion strategy that averages the detection probabilities derived from these ensembles, we achieve a more holistic and discriminative signature of AI-generated content.
- **Robust Augmentation:** To enhance robustness against “in-the-wild” degradations, we implement a hierarchical, difficulty-aware data augmentation pipeline. Combined with a Focal Loss-driven optimization and Stochastic Weight Averaging (SWA), our approach effectively bridges the distribution gap between laboratory benchmarks and real-world unconstrained imagery.

Data Collection High-quality, large-scale, and diverse datasets are fundamental to the robustness and generalizability of AI-Generated Image detection. To effectively emulate the complex, “in-the-wild” distribution of generative artifacts, we curate a comprehensive training corpus comprising millions of samples. Our data acquisition strategy is hierarchical, categorized into four primary tiers:

- **Open-Source Datasets:** To establish a solid foundation for cross-domain generalization, we integrate a diverse array of open-source resources, ranging from established academic benchmarks to large-scale repositories hosted on platforms like HuggingFace. This collection includes, but is not limited to, GenImage, WildFake, AIGIBench, CommunityForensics, and So-Fake-Set, along with representative datasets sourced from open-source communities. By aggregating these heterogeneous data sources, we ensure that our model is exposed to a wide spectrum of generation paradigms and realworld artifacts, significantly enhancing the scalability and effectiveness of our model.
- **Open-Source Generative Models:** To align with the rapid evolution of generative architectures, we construct a

substantial synthetic dataset using state-of-the-art open-source models. Our pipeline encompasses a multifaceted range of generation tasks, including Text-to-Image (T2I), Image-to-Image (I2I), image editing and inpainting, leveraging representative models such as Qwen-Image, Z-Image, and the FLUX series to capture the distinct artifacts inherent in contemporary architectures.

- **Closed-Source Commercial Models:** Given the prevalence of closedsource platforms in real-world applications, we supplement our corpus with high-fidelity samples obtained via official APIs. By integrating outputs from industry-leading engines such as Seedream, Kling, GPT-Image, and Nano-banana-pro, we effectively mitigate the distribution shift between open-source research models and proprietary commercial systems, thereby enhancing the detector’s practical deployment efficacy.
- **Challenge-Specific Datasets:** To further expand our set of training samples, we utilized the image samples provided by the competition organizers.

Methodology As illustrated in Figure 1, we propose a feature fusion architecture for AIgenerated image detection. Our approach leverages the powerful feature representation capabilities of DINOv3 backbones by employing two distinct subnetworks, each designed to process image features through an ensemble of pretrained backbones.

Specifically, the architecture comprises two independent streams: the first stream aggregates feature maps from a committee of four DINOv3 backbones, while the second stream integrates features from a separate committee of two DINOv3 backbones. Within each sub-network, the aggregated backbone features are processed through a dedicated projection layer to map them into a latent space, followed by a multi-layer perceptron (MLP) head that produces the detection probability. To derive the final prediction, we average output probabilities from both streams.

To bridge the distribution shift between controlled benchmarks and challenging “in-the-wild” imagery, we design a hierarchical, stochastic data augmentation pipeline structured by difficulty levels. Our pipeline progresses from simple, individual transformations to complex combinatorial perturbations. While the former applies individual degradations such as blur, noise, geometric shifts or compression, the latter employs multi-stage pipelines that simulate complex degradations. This hierarchical and multifaceted augmentation strategy effectively narrows the domain gap, ensuring superior detection performance even in highly unconstrained and degraded environments.

Implementation Details We initialize our framework using pretrained DINOv3 backbones and perform end-to-end fine-tuning. During training, images are randomly cropped and resized to 512×512 pixels, supplemented by our hierar-

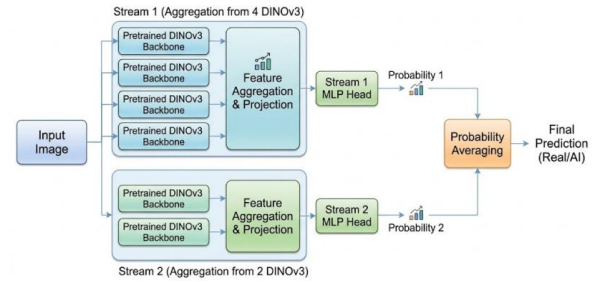


Figure 1. MICV method scheme. DINOv3-based detection framework.

chical augmentation pipeline. During inference, images are directly resized to 512×512 to preserve the global spatial context. The model is trained on 32 NVIDIA A100 GPUs for 10 epochs, with the entire procedure completing in approximately 8 hours.

- **Objective Function:** We employ Focal Loss as our primary objective function to address the potential imbalance in sample difficulty and to mitigate the dominance of easy-negative samples. The focus parameter γ is set to 2.0, and the balance parameter α is empirically set to 0.5, to ensure that the model focuses on hard-to-classify generative artifacts.
- **Optimization Strategy:** We utilize the AdamW optimizer with a weight decay of 0.02. The learning rate is initialized at 1×10^{-5} . To ensure training stability, we implement a linear warmup strategy over the first epoch, followed by a Cosine Annealing schedule to gradually decay the learning rate for the remainder of the training process. We employ Stochastic Weight Averaging (SWA) over the final epochs to aggregate model weights. This refinement yields a more stable and generalized weight configuration, which serves as our final inference model.
- **Evaluation and Refinement:** We evaluate model performance using the Area Under the Receiver Operating Characteristic curve (ROC AUC) on a dedicated validation set, which is curated by sampling 10,000 label-balanced images from the official training corpus. To ensure a robust assessment under challenging conditions, we apply a static version of the hierarchical data augmentation pipeline to this validation set to assess robustness.

Acknowledgments We sincerely thank the open-source community, whose contributions have been instrumental to the development of this framework. We are especially grateful to the creators of various AIGC detection benchmarks for providing the foundation of our study, and to the research teams behind state-of-the-art generative models for open-sourcing the architectures and tools that enabled our high-quality data synthesis.

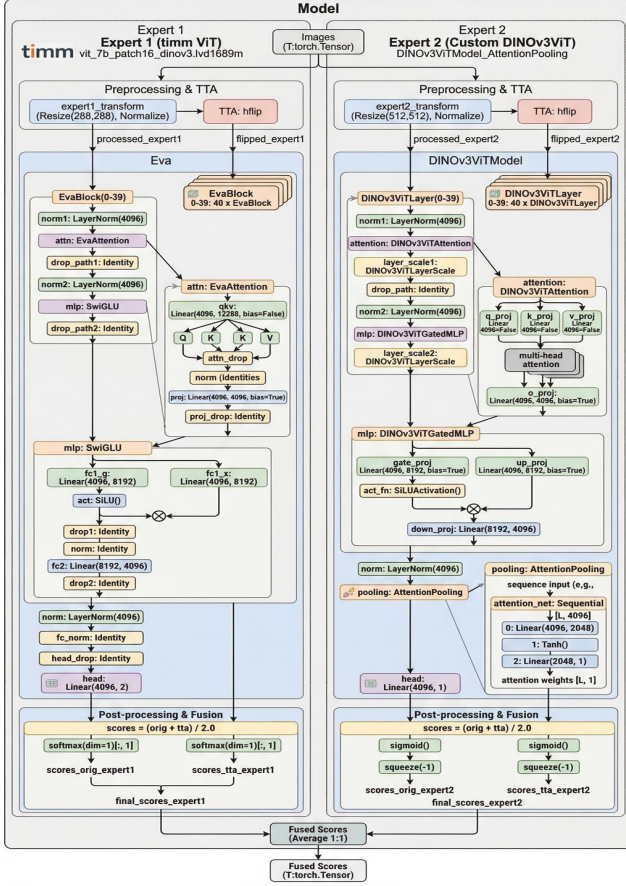


Figure 2. Ant International method scheme.

3.2. Ant International

3.2.1. Introduction

The rapid progress of text-to-image (T2I) generation has made synthetic images increasingly photorealistic, creating a pressing need for robust AI-generated image detection. Our solution for the NTIRE 2026 Robust AI-Generated Image Detection Challenge is guided by three principles:

- **Massive and diverse data:** large-scale multi-source training set with extensive augmentations.
- **Model scaling:** leveraging very large vision backbones for improved generalization.
- **Expert ensemble:** combining two complementary specialists trained with different strategies.

3.2.2. Data Preparation and Augmentation

Our training set contains approximately 1 million images from three sources:

- Official NTIRE 2026 challenge training set.
- Self-generated synthetic images from *LongCat-Image*, *Z-Image*, *Qwen-Image*, and *Qwen-Image-2512*.
- Open-source detection datasets: *Dragon* and *UniGen-Bench*.

To improve robustness, we apply a four-level offline augmentation pipeline:

- **Level 1 (Clean):** original images.
- **Level 2 (Mild):** 1–3 random distortions, severity sampled with mean 0 and std 2.5.
- **Level 3 (Moderate):** 3–6 random distortions, mean 2.5 and std 2.0.
- **Level 4 (Heavy):** fixed 6 distortions, mean 3.5, std 1.0.

3.2.3. Backbone Selection

We evaluated several large-scale vision backbones (*eva02-large*, *eva-giant*, *siglip2-giant*, *DINOv3-7B*). Experiments showed strong scaling behavior: larger models and higher input resolutions consistently improved performance and generalization. Therefore, we selected **DINOv3-7B** as the base architecture.

3.2.4. Dual-Expert Ensemble Architecture

Our final model is a dual-expert ensemble consisting of two independently fine-tuned **DINOv3-7B** models (total 14B parameters). Given an input image, both experts produce detection scores, which are aggregated into a final robust prediction. Both experts are trained with full-parameter fine-tuning on NVIDIA B200 GPUs.

3.2.5. Expert Training Strategy

Expert 1: High-Resolution Specialist

- **Data:** Levels 2, 3, and 4 only.
- **Pooling:** `attention.pooling`.
- **Input resolution:** 512×512 .
- **Online augmentation:** random horizontal flip ($p = 0.5$) + AugMix (m6-w3-d1).
- **Training setup:** 1 epoch, learning rate 1×10^{-6} , Model EMA, AMP, no dropout.

Expert 2: Robustness-Focused Specialist

- **Data:** Levels 1, 2, 3, and 4.
- **Pooling:** `first.token.pooling`.
- **Input resolution:** 288×288 .
- **Training duration:** 10 epochs.

3.2.6. Inference

Inference consists of two steps:

1. **Test-time augmentation (TTA):** each expert predicts on multiple augmented variants of each test image.
2. **Weighted ensembling:** TTA outputs are aggregated per expert, then combined via weighted averaging across experts.

On a single NVIDIA A100 GPU, the full pipeline runs at approximately **2.21 images/s** with **78.25 GB** VRAM usage.

3.2.7. Conclusion

Our NTIRE 2026 solution shows that robust AI-generated image detection benefits strongly from:

- model scaling (DINOv3-7B),
- large and diverse augmented data,
- and complementary dual-expert ensembling.

By combining a high-resolution specialist and a robustness-focused specialist, we achieve strong generalization under real-world image transformations.

3.3. TeleAI-TeleGuard. TeleAI Strategy for Robust AI-Generated Image Detection in the Wild

3.3.1. Proposed Method

We propose a LoRA-based Pairwise Training (LPT) strategy to achieve robust detection of AI-generated images (AIGI). As shown in Figure 3, this strategy consists of three components: foundation-model fine-tuning, distortion simulation, and pairwise optimization.

In addition to the distortions provided by the organizer, after inspecting the validation set we introduce three extra distortion types—*Speckle Noise*, *Color Cast*, and *Organic Moire*—which better match the target data distribution. We also increase distortion severity by setting the mean of the Gaussian distribution to 3.

The foundation model in LPT is EVA-CLIP [70]. Following LoRA-style fine-tuning [32], we adapt the linear layers in the multi-head self-attention (MHSA) and feed-forward network (FFN) of each visual transformer block.

To improve robustness without degrading clean-sample performance, we adopt pairwise optimization by jointly feeding clean images and their corresponding distorted versions in each batch. Features extracted from distorted images are corrected by an additional feed-forward network. The overall training loss is:

$$\mathcal{L} = \mathcal{L}_{CE}(\mathbf{x}, \mathbf{y}) + \alpha \cdot \mathcal{L}_{KL}(\mathbf{x}, \hat{\mathbf{x}}) + \beta \cdot \mathcal{L}_{MSE}(\mathbf{f}_x, \mathbf{f}'_x), \quad (1)$$

where $\mathbf{x}$, $\hat{\mathbf{x}}$, $\mathbf{f}_x$, and $\mathbf{f}'_x$ denote clean samples, distorted samples, clean-sample features, and corrected distorted-sample features, respectively. $\mathcal{L}_{CE}$, $\mathcal{L}_{KL}$, and $\mathcal{L}_{MSE}$ are cross-entropy, KL divergence, and mean squared error losses. We set $\alpha = 0.5$ and $\beta = 0.25$.

Experiments are conducted on eight NVIDIA A800 GPUs for 5 epochs. We use AdamW [48] with an initial learning rate of 2×10^{-4} and a cosine annealing scheduler [47]. To further improve generalization, we additionally include the So-Fake [33] and Chameleon [86] datasets during training.

3.4. INTSIG

3.4.1. Model Training Strategies

Overview. We develop five complementary detectors in a staged pipeline. Model 1 is a full fine-tuning baseline on DINOv3-Huge. Model 2 continues training with expanded data. Model 3 further increases robustness with stronger distortion augmentation. Model 4 introduces a

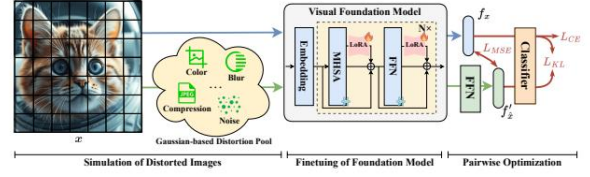


Figure 3. TeleAI-TeleGuard method scheme. **Pipeline of LPT.** To achieve robust detection, the framework includes distortion simulation, foundation-model fine-tuning, and pairwise optimization.

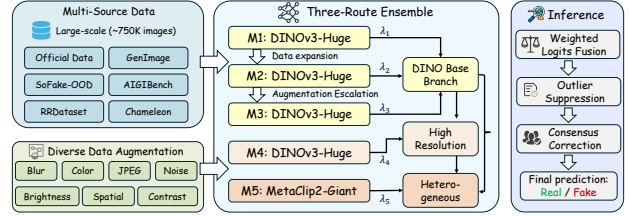


Figure 4. INTSIG method scheme.

high-resolution branch (448×448). Model 5 replaces the backbone with MetaCLIP2 Giant and applies partial fine-tuning. This design balances in-domain accuracy and OOD robustness.

Model 1: Baseline. Model 1 uses DINOv3-Huge with an MLP head:

$$1280 \rightarrow 256 \xrightarrow{\text{ReLU+Dropout}(0.1)} 2. \quad (2)$$

We use AdamW (betas (0.9, 0.999)) with separate parameter groups: backbone $\text{lr} = 2 \times 10^{-5}$, weight decay = 0.05; head $\text{lr} = 5 \times 10^{-4}$, weight decay = 0.01. The learning-rate schedule is 10% warmup followed by cosine annealing, stepped per batch. Cross-entropy is used as the training objective, and class imbalance is handled by a distributed weighted sampler. Training data includes the official training set, SoFake-OOD, and RRDataset (10% validation split). Inputs are preprocessed with random resized crop (256×256 , bicubic, scale (0.08, 1.0), ratio (0.75, 1.3333)), random horizontal flip, and color jitter. Distortion augmentation is enabled with probability 0.5 (up to 3 distortions, 3 levels). Training runs for 20 epochs on $8 \times \text{H800}$ (DDP), with per-GPU batch size 16/8 (train/val).

Model 2: Incremental training with dataset expansion.

Starting from Model 1, we reduce learning rates (backbone 5×10^{-6} , head 5×10^{-5}) and expand training data to: official set + SoFake-OOD + RRDataset + Chameleon + 50% GenImage_val + AIGIBench_test. Other settings are unchanged.

Model	Input	Core change	Weight
M1	256 ²	DINOv3-Huge baseline	0.3675
M2	256 ²	Incremental data expansion	0.0735
M3	256 ²	Stronger distortion augmentation	0.0490
M4	448 ²	High-resolution branch	0.2100
M5	378 ²	MetaCLIP2 Giant backbone	0.3000

Table 1. Summary of training paths and ensemble weights.

Model 3: Incremental training with enhanced augmentation. Model 3 continues from Model 2 while increasing distortion intensity to 5 distortions and 5 levels (augmentation probability remains 0.5). Other settings are unchanged.

Model 4: High-resolution path. Model 4 uses 448×448 inputs to capture finer artifacts. AdamW is retained with 5% warmup and cosine annealing. Learning rates are set to 5×10^{-5} (backbone) and 1×10^{-3} (head). Training data is the official set only, with a 1% validation split. Preprocessing follows the same policy as Model 1 but at 448×448 resolution. Distortion augmentation uses probability 0.5, with up to 3 distortions and 5 levels.

Model 5: Backbone replacement. Model 5 adopts MetaCLIP2 Giant. We partially fine-tune all LayerNorm parameters and the full parameters of the last two layers. The classifier is:

$$\begin{aligned} \text{MLP1: } &1664 \rightarrow 2048 \rightarrow 512, \\ \text{MLP2: } &512 \rightarrow 256 \rightarrow 2. \end{aligned} \quad (3)$$

Training uses the official set, learning rate 10^{-4} , and FocalLoss. Inputs are resized to 378×378 ; training includes random flip, random rotation, and random augmentation. Distortion augmentation is lighter (probability 0.2, 3 distortions, 3 levels).

3.4.2. Model Ensemble Strategy

Weighted hierarchical fusion. The final logit is computed as:

$$\begin{aligned} \text{Final_Logit} = &0.7 \left[0.7(0.75M1 + 0.15M2 + 0.1M3) \right. \\ &\left. + 0.3M4 \right] + 0.3M5. \end{aligned} \quad (4)$$

Horizontal-flip TTA is applied to M3 and M4.

Dual-gating correction. We define directional confidence:

$$\text{diff} = \text{logit}_1 - \text{logit}_0. \quad (5)$$

Gate-1 (strong consensus correction) is triggered when M4 and M5 strongly agree ($|M4_{\text{diff}}| \geq 8$, $|M5_{\text{diff}}| \geq 3$), but

Dimension	Technical solution	Effect
GPU parallelism	5 independent streams	Full model overlap
Compute precision	AMP for M1–M4	~30% throughput gain
Data loading	Async I/O + prefetch=4	Reduced wait time
Pipeline	CPU/GPU overlap	Higher utilization

Table 2. Inference-time engineering optimizations.

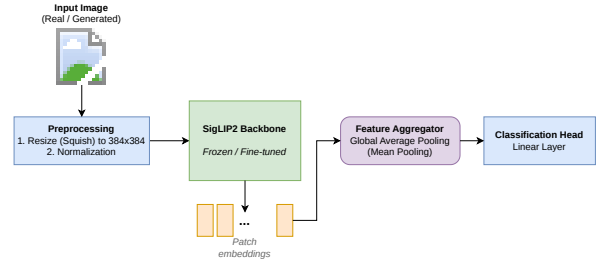


Figure 5. Vincentlc method scheme.

the fused output has the opposite direction. We then shift the final logit by 2.5 toward the M4/M5 direction. *Gate-2 (anomaly suppression)* is triggered when at least 3 models in $\{M1, M2, M3, M5\}$ agree while M4 disagrees; M4 is excluded and remaining weights are renormalized.

3.4.3. Engineering Optimization

Parallel inference. Inference runs with one CUDA stream per model, CPU-side multi-threaded preprocessing, and AMP for M1–M4.

I/O and overlap. We use asynchronous data loading (num_workers=4) and prefetching (prefetch=4), overlapping CPU preprocessing with GPU inference to reduce idle time.

3.5. Vincentlc. Robust AI-Generated Image Detection via SigLIP2-Giant and Perturbation-Aware Training

3.5.1. Overview

We propose a simple yet effective framework for detecting AI-generated images in the wild. Our method combines a pre-trained vision-language backbone with a lightweight linear classification head. To improve robustness under severe real-world distribution shifts, we explicitly integrate the official image distortion pipeline used in challenging evaluation protocols [37] into training, forcing the network to learn perturbation-invariant representations.

3.5.2. Model Architecture

Backbone Feature Extractor. We experimented with multiple scales of the SigLIP2 family and found that the

Method	Open Test		Hidden Test		Average	
	ROC AUC	Rob. ROC AUC	ROC AUC	Rob. ROC AUC	ROC AUC	Rob. ROC AUC
MICV	0.9978	0.9738	<u>0.9969</u>	<u>0.9707</u>	0.9974	0.9723
Ant International	<u>0.9973</u>	<u>0.9731</u>	0.9971	0.9711	<u>0.9972</u>	<u>0.9721</u>
TeleAI-TeleGuard	0.9762	0.9215	0.9809	0.9286	0.9786	0.9251
INTSIG	0.9810	0.9090	0.9895	0.9171	0.9897	0.9130
vincentlc	0.9497	0.8633	0.9557	0.8828	0.9527	0.8730
UESTC	0.9693	0.8558	0.9764	0.8800	0.9729	0.8679
Reagvis Labs	0.9423	0.8474	0.9481	0.8733	0.9452	0.8603
PSU	0.9132	0.8334	0.9322	0.8483	0.9227	0.8408
Shallow Real	0.9954	0.8302	0.9951	0.8370	0.9953	0.8336

Table 3. Final results and rankings of the top-9 teams. Performance is reported on the open and hidden test sets using both clean ROC-AUC and Robust ROC-AUC. The final ranking is determined by the average Robust ROC-AUC across both splits.

largest variant, siglip2-giant-opt-patch16-384, achieves the strongest performance. For downstream binary classification, we extract the patch-token sequence from the final hidden layer of the pre-trained SigLIP2 encoder [23].

Feature Aggregation and Classification Head. We evaluated several aggregation strategies, including CLS-token extraction, attention pooling, and multi-layer feature concatenation. Empirically, global average pooling over all final-layer patch tokens yields the most robust and stable results. The pooled feature vector is then fed into a single linear layer to produce real/fake logits.

3.5.3. Robust Data Augmentation

Spatial Transforms (“Squish” Strategy). Instead of random resized cropping (which may remove localized forensic cues), we apply a “squish” strategy: all images are directly resized to 384×384 (ignoring aspect ratio), followed by random horizontal flipping. This preserves full image content while maintaining training efficiency.

Competition-Specific Distortions. We wrap the official `distort_images` function into the PyTorch transform pipeline. In our best run, we set `distortion_prob=1.0`, so every training image is distorted, with up to three random post-processing operations sampled at multiple severity levels (`num_levels=5`). This aggressive augmentation acts as a strong regularizer and is a key driver of our Robust ROC AUC improvements.

4. Results

Table 3 presents the final scores and rankings of the top-9 participating teams. All submissions were verified by re-running the provided models in the official evaluation environment to ensure consistency with the public leaderboard.

The final ranking is determined based on the average Robust ROC-AUC across the open and hidden test datasets.

For completeness, we also report the average clean ROC-AUC, as well as all individual metrics, including clean and robust ROC-AUC on both dataset splits.

Overall, the top-performing methods achieve ROC AUC values exceeding 0.99. The **MICV** team ranks first according to the primary metric, achieving the highest average Robust ROC-AUC of 0.9723. Their method also delivers the best average clean ROC-AUC (0.9974), indicating strong performance under both standard and robustness-aware evaluations. **Ant International** closely follows in second place, achieving the best performance on the hidden test set for both clean and robust ROC-AUC (0.9971 and 0.9711, respectively). This demonstrates excellent generalization to unseen data. The small margin between the top-2 teams suggests that both approaches are competitive and approach the upper bound of performance on this benchmark. A clear performance gap is observed between the top-2 teams and the remaining participants. Methods such as **TeleAI-TeleGuard** and **INTSIG** achieve strong clean ROC AUC scores (around 0.98–0.99), but show noticeably lower robustness, with Robust ROC-AUC dropping to approximately 0.91–0.93. This indicates increased sensitivity to challenging conditions and highlights robustness as a key differentiating factor. The lower-ranked methods, including **vincentlc** and **UESTC**, exhibit further degradation in both clean and robust metrics, particularly in the robustness setting, where performance falls below 0.88.

5. Conclusion

In the NTIRE 2026 Robust AI-Generated Image Detection in the Wild, 500+ registered participants competed to develop state-of-the-art deepfake detection methods for a novel dataset of 250,000+ images. Most final solutions used Expert-base architecture with several different models and use Transformer-based architectures. The results suggest that, although performance has improved substantially, the problem is not yet solved, and there remains room for advances in design, training strategies and data curation.

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7. Teams and Affiliations

7.1. NTIRE 2026 Challenge Organizing team

Members:

Aleksandr Gushchin¹ (alexander.gushchin@graphics.cs.msu.ru),
Khaled Abud¹ (khaled.abud@graphics.cs.msu.ru),
Ekaterina Shumitskaya¹ (ekaterina.shumitskaya@graphics.cs.msu.ru),
Artem Filippov² (artyom.filippov@graphics.cs.msu.ru),
Georgii Bychkov¹ (georgy.bychkov@graphics.cs.msu.ru),
Sergey Lavrushkin¹ (sergey.lavrushkin@graphics.cs.msu.ru),
Mikhail Erofeev⁵ (mikhail@erofeev.pw),
Anastasia Antsiferova¹ (Aantsiferova@graphics.cs.msu.ru),
Changsheng Chen³ (cschen@smbu.edu.cn),
Shunquan Tan³ (tansq@smbu.edu.cn),
Radu Timofte⁴ (radu.timofte@uni-wuerzburg.de),
Dmitry Vatolin¹ (dmitriy@graphics.cs.msu.ru)

Affiliations:

¹ MSU Institute for Artificial Intelligence

² Lomonosov Moscow State University

³ Shenzhen MSU-BIT University

⁴ University of Würzburg

⁵ Independent Researcher

7.2. MICV

Members:

Chuanbiao Song¹ (songchuanbiao.scb@antgroup.com),
Zijian Yu¹ (yuzijian.yzj@antgroup.com),
Hao Tan¹ (tanhao2023@ia.ac.cn),
Jun Lan¹ (yelan.lj@antgroup.com)

Affiliations:

¹ Ant Group, China

7.3. Ant International

Members:

Zhiqiang Yang¹ (874125760@qq.com),
Yongwei Tang¹ (1598521844@qq.com),

Zhiqiang Wu¹ (1753940858@qq.com),
Jia Wen Seow¹ (sscarlettss0111@gmail.com),
Hong Vin Koay¹ (koayhv@gmail.com),
Haodong Ren¹ (renhaodong.rhd@ant-intl.com),
Feng Xu¹ (fuyu.xf@ant-intl.com),
Shuai Chen¹ (shuai.cs@ant-intl.com)

Affiliations:

¹ Ant International, Singapore

7.4. TeleAI-TeleGuard

Members:

Ruiyang Xia¹ (ryon@stu.xidian.edu.cn),
Qi Zhang¹ (zhangq139@chinatelecom.cn),
Yaowen Xu¹ (xuyw1@chinatelecom.cn),
Zhaofan Zou¹ (zouzhf41@chinatelecom.cn),
Hao Sun¹ (sunh10@chinatelecom.cn)

Affiliations:

¹ TeleAI-TeleGuard, China

7.5. INTSIG

Members: Fei Wu^{1,2} (wu_fei@sjtu.edu.cn),
Dagong Lu¹ (dagong_lu@intsig.net),
Mufeng Yao¹ (mufeng_yao@intsig.net),
Xinlei Xu¹ (xinlei_xu@intsig.net),
Fengjun Guo¹ (fengjun_guo@intsig.net)

Affiliations:

¹ IntSig Information Co. Ltd, Shanghai, China

² Shanghai Jiao Tong University, Shanghai, China

7.6. Vincentlc

Members:

Cong Luo (1486891983@qq.com)

7.7. Reagvis Labs

Members:

Hardik Sharma¹ (d25092@students.iitmandi.ac.in),
Aashish Negi¹ (dd25011@students.iitmandi.ac.in),
Prateek Shaily¹ (prateekshaily820@gmail.com),
Jayant Kumar¹ (aryanjay2k05@gmail.com),
Sachin Chaudhary¹ (sachin.chaudhary@ddn.upes.ac.in),
Akshay Dudhane¹ (akshay.dudhane@mbzuai.ac.ae),
Praful Hambarde¹ (praful@iitmandi.ac.in),
Amit Shukla¹ (amitshukla@iitmandi.ac.in)

Affiliations:

¹ IIT Mandi, Reagvis Labs, Mandi, India

7.8. UESTC

Members:

Zhilin Tu¹ (202521080234@std.uestc.edu.cn),
Fengpeng Li¹ (fengpeng.li@connect.umac.mo),
Jiamin Zhang¹ (18482160071@163.com),
Jianwei Fei¹ (fei_jianwei@163.com),

Kemou Li¹ (yc47912@umac.mo),
 Haiwei Wu¹ (haiweiwu@uestc.edu.cn)

Affiliations:

¹ University of Electronic Science and Technology of
 China, China

7.9. PSU

Members:

Bilel Benjdira¹ (bbenjdira@psu.edu.sa),
 Anas M. Ali¹ (aaboessa@psu.edu.sa),
 Wadii Boulila¹ (wboulila@psu.edu.sa)

Affiliations:

¹ Prince Sultan University, Robotics and Internet-of-Things
 Laboratory, Riyadh, Saudi Arabia

7.10. ShallowReal

Members:

Chenfan Qu¹ (202221012612@mail.scut.edu.cn),
 Junchi Li² (hongge568@126.com)

Affiliations:

¹ South China University of Technology, Guangzhou, China
² Zhejiang University, Hangzhou, China

8. Challenge Methods and Teams (Continued)

**8.1. Reagvis Labs. RAPID: Robust AI-Generated
 Prototype Image Detection**

We propose RAPID, a cascaded six-model ensemble for robust AI-generated image detection under realistic degradations. The system combines four complementary representation families: CLIP [62], SigLIP [90], SRM-style forensic residual modeling [20], and EVA-02 masked-image pre-training [18]. Each branch is fine-tuned with parameter-efficient adaptation (mainly LoRA [32]), and final prediction is obtained by staged logit-space fusion. To keep memory low, models are executed sequentially; peak VRAM remains below 4 GB.

8.1.1. Architecture

RAPID uses six detectors in a progressive refinement cascade. The main semantic branch (G4) is CLIP ViT-L/14 with LoRA on attention and MLP projections, followed by a GAPL prototype-attention head that compares normalized embeddings against learned forensic prototypes. A high-resolution SigLIP-v2 branch (384×384) shares the same GAPL idea and improves sensitivity to fine local artifacts. In parallel, an SRM+Bayar ForensicCNN branch processes raw pixel inputs without normalization, explicitly modeling residual-domain inconsistencies. Two EVA-02 checkpoints provide complementary MIM-based cues: the original LoRA model and a retrained fixed-initialization variant with improved calibration under degradation. Finally, G4_v2 (continued training from G4 EMA) performs last-step correction on difficult modern generations.

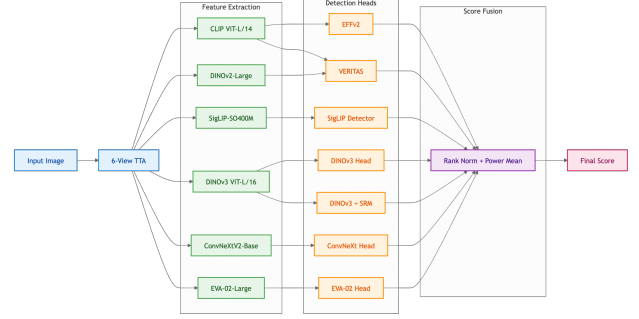


Figure 6. Reagvis Labs method scheme. Overview of RAPID with staged logit fusion.

8.1.2. Logit-Space Cascade Fusion

All branches output probabilities in $[0, 1]$. We combine them in logit space, where additive mixing is better calibrated than direct probability averaging.

$$\ell(p) = \log \frac{p}{1-p}. \quad (6)$$

$$\begin{aligned} s_{1a} &= \sigma(0.50L_{G4} + 0.35L_S + 0.15L_R), \\ b &= \sigma(0.80L_{1a} + 0.20L_E), \\ s_2 &= \sigma(0.85L_b + 0.15L_{E_f}), \\ f &= \sigma(0.89L_2 + 0.11L_{G4v2}). \end{aligned} \quad (7)$$

Here, $L_{G4} = \ell(G4)$, $L_S = \ell(\text{SigLIP})$, $L_R = \ell(\text{SRM})$, $L_E = \ell(\text{EVA02})$, $L_{E_f} = \ell(\text{EVA02}_{\text{fixed}})$, $L_{1a} = \ell(s_{1a})$, $L_b = \ell(b)$, $L_2 = \ell(s_2)$, and $L_{G4v2} = \ell(G4_{v2})$.

This cascade was built incrementally through validation ablations; each stage was retained only if it improved robust AUC.

8.1.3. Inference-Time Robustness

We apply 8-view TTA per model, mixing geometric and degradation-aware transforms (flip, multi-scale center crops, corner crops, blur, and JPEG perturbation). For sigmoid heads, we average logits across views and then apply sigmoid; for softmax heads, we average class probabilities. SigLIP uses resolution-matched 384px variants of the same views, while SRM uses bicubic resizing on raw $[0, 1]$ inputs.

8.1.4. Training Protocol

All branches are trained on the official NTIRE 2026 data using AdamW with cosine decay and warmup, EMA checkpointing, and mixed precision where stable. Augmentations include random crop/flip, JPEG recompression, Gaussian blur/noise, and random resizing. Transformer branches use LoRA (typically $r = 32$), while G4_v2 is obtained by low-LR continuation from the G4 EMA checkpoint with SAM and CutMix for harder-sample generalization.

Model	Time (ms)	Peak VRAM (GB)	Fine-tuning
G4 (CLIP+GAPL)	~37	2.6	LoRA $r=32$ (attn+MLP)
SigLIP-v2	~26	1.8	LoRA $r=32$ (384px)
SRM ForensicCNN	~5	0.1	full training (FP32)
EVA-02 (original)	~25	1.3	LoRA $r=32$ + SwiGLU
EVA-02 (fixed)	~25	1.3	LoRA $r=32$ retrain
G4_v2 (CLIP+GAPL)	~37	2.6	LoRA continuation

Table 4. Per-branch inference cost and memory. Sequential loading keeps total peak memory below 4 GB.

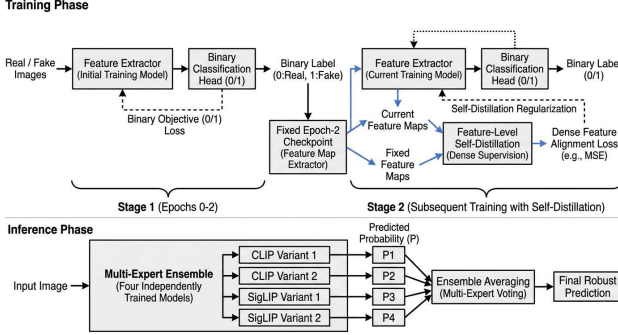


Figure 7. UESTC method scheme.

8.1.5. Runtime and Memory

Table 4 reports per-model latency and memory on an H100.

8.1.6. Design Rationale

RAPID is built around complementarity: semantic and language-aligned features (CLIP/SigLIP), residual forensic cues (SRM), and MIM-driven representations (EVA-02) fail on different samples, making staged logit blending consistently stronger than any single branch. Prototype-attention heads improve interpretability by comparing each image to learned real/fake forensic anchors, while degradation-aware TTA stabilizes predictions under compression, blur, and scale shifts commonly observed in the wild.

8.2. UESTC

Overview. Our final detector is a multi-expert ensemble composed of four Vision Transformer backbones: two CLIP ViT-L/14 models [62] and two SigLIP So400M Patch14-384 models [90]. For each backbone, we attach a lightweight classification head that maps high-dimensional feature representations to binary predictions. Because the backbones have different native input configurations, we use backbone-specific preprocessing and standardize inputs to 224×224 for CLIP and 384×384 for SigLIP. Despite using four experts, inference remains efficient, with a peak GPU memory footprint of approximately 10 GB.

Backbone selection. To identify suitable feature extractors, we conduct a large-scale empirical study over diverse backbones, including CLIP [62], SigLIP [90], MoCo [28], Swin Transformer [45], ConvNeXt [46], DINO [8], CoCa [88], and BEiT [5]. We compare representation quality, computational efficiency, and robustness to synthetic artifacts. Empirically, CLIP and SigLIP provide the best trade-off between discriminative capability and cross-domain generalization, and therefore form the basis of the final system.

Data strategy. We improve robustness through two complementary components: training-set expansion and comprehensive degradation modeling. First, we expand the official training set with diverse synthesis pipelines, including state-of-the-art diffusion generators [29, 64], reconstruction pipelines, domain-transfer methods, and targeted adversarial attacks. This expansion introduces both realistic AI-generated samples and challenging hard negatives, exposing the detector to broader failure modes and emerging forgery artifacts. Second, we extend the official augmentation scripts with a wider array of degradation operators, covering complex blurs, noise patterns, compression artifacts, and advanced optical/digital distortions. These realistic perturbations encourage degradation-invariant representations and significantly improve robustness under distribution shifts caused by imaging conditions, transmission errors, and post-processing pipelines.

Training paradigm. We treat deepfake detection as more than a vanilla binary classification task and adopt a two-stage training strategy. In Stage 1, each model is trained for two epochs with a standard binary objective (real vs. generated). In Stage 2, we introduce feature-level self-distillation with dense supervision: intermediate feature maps from the epoch-2 checkpoint are used as dense targets, and the model is explicitly optimized to align current representations with these distilled features. This dense alignment regularizes feature geometry, reduces overfitting to the competition distribution, and improves generalization to unseen generation methods and real-world domains.

Inference. At inference time, we average the predicted probabilities from four independently trained experts (two CLIP variants and two SigLIP variants) to produce the final robust prediction.

8.3. PSU. PRISM: Paradigm-diverse Representation Integration for Synthesis-artifact Manifold Detection

8.3.1. Method Description

PRISM is a heterogeneous ensemble detector for AI-generated images, designed for robustness under JPEG

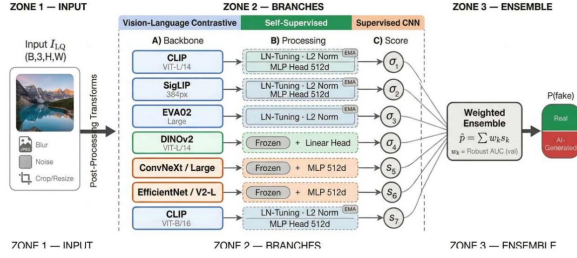


Figure 8. PSU method scheme.

compression, blur, noise, rescaling, and cropping. Our core hypothesis is that no single pre-training objective encodes the full forensic artifact manifold [14, 78]: contrastive objectives capture semantic inconsistencies; self-supervised patch objectives preserve texture discontinuities; supervised CNNs encode low-frequency spectral anomalies. Robust detection therefore requires *explicit paradigm diversity*.

Prior work. Wang *et al.* [78] showed CNN detectors generalize across GAN generators with augmentation, yet collapse under JPEG re-compression [24]. Ojha *et al.* [55] froze CLIP and trained a linear probe, achieving strong cross-generator generalization. PRISM extends this by (i) covering three paradigms beyond CLIP, (ii) paradigm-specific fine-tuning, and (iii) weighting by *robust AUC* rather than clean accuracy.

Encoder pool. We instantiate $K=7$ encoders $\{\phi_k\}_{k=1}^K$ across three paradigms (Table 5). For vision-language (VL) encoders (CLIP [62], SigLIP [90], EVA02 [18]), features are ℓ_2 -normalized onto the unit hypersphere [79]:

$$\hat{z}_k = \frac{z_k}{\|z_k\|_2} \in \mathcal{S}^{d_k-1},$$

preserving the contrastive geometry. DINOv2 [56], ConvNeXt [46], and EfficientNet-V2 [72] remain frozen; their features are passed directly to the classification head.

Paradigm-aware tuning. VL encoders undergo *Layer-Norm tuning* [82]: only LayerNorm scale/shift (about 0.03% of weights) are updated, preventing catastrophic forgetting while adapting internal normalization to the forensic domain. All models use EMA ($\alpha=0.9995$) for checkpoint stabilization and calibrated posteriors. VL heads are two-layer MLPs ($h=512$); DINOv2 uses a linear probe. Training uses label-smoothed BCE ($\varepsilon=0.05$), AdamW ($\lambda=0.01$), cosine LR schedule, and fp16.

Robust ensemble. Each model is scored on held-out validation data under all degradation types, yielding per-model

Table 5. PRISM encoder pool. d : feature dimension. LN: LayerNorm-only tuning. F: frozen encoder.

Encoder	Paradigm	d	Tune	Head
CLIP ViT-L/14	VL-Contrastive	1024	LN	MLP-512
SigLIP ViT-L/16	VL-Contrastive	1024	LN	MLP-512
EVA02-Large	VL-Contrastive	1024	LN	MLP-512
DINOv2 ViT-L/14	Self-Supervised	1024	F	Linear
ConvNeXt-Large	Supervised CNN	1536	F	MLP-512
EfficientNet-V2-L	Supervised CNN	1280	F	MLP-512
CLIP ViT-B/16	VL-Contrastive	512	LN	MLP-256

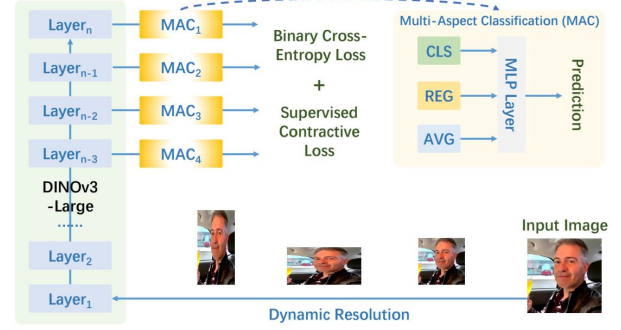


Figure 9. Shallow Real method scheme. Overall pipeline of the proposed model.

robust AUC A_k^{rob} . Normalized weights

$$w_k = \frac{A_k^{\text{rob}}}{\sum_j A_j^{\text{rob}}}$$

drive the ensemble. The final prediction with horizontal-flip TTA is

$$\hat{p} = \frac{1}{2} \left[\sum_k w_k p_k(x) + \sum_k w_k p_k(\mathcal{F}(x)) \right]. \quad (8)$$

Early stopping maximizes

$$\mathcal{M} = 0.7A^{\text{rob}} + 0.3A^{\text{clean}}.$$

8.4. Shallow Real

8.4.1. Method Description

As shown in Fig. 9, we formulate robust AI-generated image detection as a binary classification task. Our model uses a DINOv3-Large backbone (following the DINO line of self-supervised visual encoders [56]), fine-tuned with Low-Rank Adaptation (LoRA) [32] using rank $r = 32$ and scaling factor $\alpha = 64$.

Final prediction is produced by a Multi-Aspect Classification (MAC) head operating on DINOv3 features. Specifically, the head aggregates: (i) one [CLS] token, (ii)

four [REG] tokens, and (iii) one [AVG] token computed as the average of the remaining patch tokens. These six 1024-dimensional vectors are concatenated into a 6144-dimensional representation and passed to a two-layer MLP with an intermediate ReLU activation to output the binary score.

During training, we use three key techniques:

1. **Dynamic resolution.** Input images are resized to random resolutions, where both height and width are sampled from [384, 1152] (multiples of patch size 16), with a randomly selected interpolation method.
2. **Deep supervision.** Auxiliary classification heads are attached to the last four layers of the DINOv3 backbone. Training loss is computed for each auxiliary output, while only the final-layer output is used at inference time. Each auxiliary head uses dropout with rate 0.2 in its fully connected layer.
3. **Metric learning objective.** A supervised contrastive loss [35] is added as an auxiliary objective alongside the primary binary cross-entropy loss, improving feature discrimination.

The model is trained for 10 epochs using AdamW [48] with a linear learning-rate decay from 10^{-4} to 0. Training uses the union of the official training set and the CommunityForensics dataset, with batch size 32 and the official data-augmentation pipeline.

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Split	# of Images	Real/Fake Ratio	Labels Provided	Generator Models		Transformations	
Train	~277K	~1:1.77	Yes	20 Models:		12 Transformations: (Provided as distortion pipeline)	
				• YOSO PixArt-512 • PixArt-α • PixArt-Σ • Kandinsky 2 • Kandinsky 3 • Kolos • OmniGen • OmniGen 2 • Stable Diffusion 1.4 • Stable Diffusion 1.5 • Stable	• Diffusion 2.1 • Stable Diffusion XL 1.0 • SDXL Lightning • SDXL Turbo • Janus Pro 7B • Infinity 2B • Infinity 8B • Ovis Image • DeepFloyd IF • FLUX.1 • Kontext Dev	• Gaussian Blur • Lens Blur • Color Shift • Color Saturation • JPEG Compression • White Noise • Impulse Noise • Brightness	• Increase • Brightness • Decrease • Color Jitter • Color Quantization • Linear Contrast Change
Validation	10K	1:1	Yes (After Completion)	9 Models:		19 Transformations:	
				• FLUX.1 Kontext Dev • SDXL Turbo • FLUX.1 Dev • Playground v2.5 • Lumina Image 2.0	• Qwen Image • Stable Diffusion 3 Medium • Ideogram Turbo[†] • ImageGen-4 Fast[†]	• Gaussian Blur • Lens Blur • Color Shift • Color Saturation • JPEG Compression • White Noise • Impulse Noise • Brightness Increase • Brightness Decrease • Color Jitter • Color Quantization	• Linear Contrast Change • Motion Blur • Multiplicative Noise • Pixelation • RGB Channel Shift • Random Crop • Random Aspect Crop • Downscale
Validation Hard	2.5K	1:1	Yes (After Completion)	7 Models:		19 Transformations:	
				• Playground v2.5 • SDXL Turbo • HiDream • FLUX.1 Schnell	• Stable Diffusion 3.5 Large Turbo • Nano Banana[†] • Seedream 4[†]	• Gaussian Blur • Lens Blur • JPEG Compression • White Noise • Impulse Noise • Color Quantization • Multiplicative Noise • RGB Channel Shift • Random Crop • Random Aspect Crop • Neural Image Compression (JPEG AI) • Random Tone	• Curve • CLAHE • ISO Noise • Perspective Transform • Multiple Compressions (JPEG) • Multiple Compressions (JPEG + JPEG AI) • Watermark • Attack (Adv. Embedding, CLIP/ResNet) • Downscale

Table 6. Detailed information on Train, Validation, and Validation Hard dataset splits. [†] denotes a proprietary generation model used via API.

Split	# of Images	Real/Fake Ratio	Labels Provided	Generator Models	Transformations
Test (Public)	2.5K	1:1	Yes (After Completion)	10 Models: • HiDREAM • FLUX.1 Schnell • Stable Diffusion 3.5 Large • FLUX Krea • Z-Image Turbo • Nano Banana	22 Transformations: • Color Saturation (JPEG) • Brightness Increase • Lens Blur • JPEG Compression • Impulse Noise • RGB Channel Shift • Random Crop • Random Aspect Crop • Neural Image Compression (JPEG AI) • Random Tone Curve • CLAHE • ISO Noise • Perspective Transform • Multiple Compressions • Multiple Compressions (JPEG + JPEG AI) • Watermark Attack (Adv. Embedding, CLIP/ResNet) • JPEG 2000 • Watermark Attack (WMForger) • Neural Image Compression (Cheng2020) • Shot Noise • Downscale • Invisible Watermark Insertion (1 of 6 algorithms)
Test (Private)	2.5K	~1:1	No	10 Models: • HiDREAM • Stable Diffusion 3.5 Large Turbo Dev SRPO • Z-Image Turbo • Kandinsky 5 • Nano Banana 2[†] • GPT Image 1.5[†] • ImageGen-4 Ultra[†] • Seedream 5 Lite[†] • Grok Imagine Image[†]	24 Transformations: • Color Saturation (JPEG + JPEG AI) • Brightness Increase • Lens Blur • JPEG Compression • Impulse Noise • RGB Channel Shift • Random Crop • Random Aspect Crop • Neural Image Compression (JPEG AI) • Random Tone Curve • CLAHE • ISO Noise • Perspective Transform • Multiple Compressions (JPEG) • Multiple Compressions • Multiple Compressions (JPEG + JPEG AI) • Multiple Compressions (JPEG + JPEG 2000) • Watermark Attack (Adv. Embedding, CLIP/ResNet) • JPEG 2000 • Watermark Attack (WMForger) • Neural Image Compression (Cheng2020) • Shot Noise • Glass Blur • Downscale • Invisible Watermark Insertion (1 of 7 algorithms)

Table 7. Detailed information on Test (public) and Test (private) dataset splits. [†] denotes a proprietary generation model used via API.